

Task 1 — Project Report

Build & Evaluate a Linear Regression Model

California Housing Dataset — House Price Predictor

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Domain	Artificial Intelligence & Machine Learning
Date	June 2026
Dataset	California Housing (20,640 samples)
Algorithm	Linear Regression (scikit-learn)

Objective

Train a Linear Regression model on the California Housing dataset, covering the full ML workflow: data loading, exploration,

Skills Demonstrated

- Python programming with pandas, NumPy, scikit-learn, matplotlib, seaborn
- Exploratory Data Analysis (EDA) — distributions, correlations, missing value checks
- Data preprocessing — train/test split, StandardScaler feature normalization
- Model training using scikit-learn LinearRegression
- Model evaluation using MAE, RMSE, and R^2 metrics
- Data visualization — scatter plots, heatmaps, residual analysis

1. Dataset Overview

The California Housing dataset contains 20,640 records from the 1990 California census. Each row represents one block group (neighbourhood). The dataset has 8 input features and 1 target variable (MedHouseVal — median house value in \$100k units). No missing values were found — the dataset is clean and ready for modelling.

Feature	Description	Type
MedInc	Median income of block group (\$10k)	Numerical
HouseAge	Median house age in years	Numerical
AveRooms	Average number of rooms per household	Numerical
AveBedrms	Average bedrooms per household	Numerical
Population	Total population of block group	Numerical
AveOccup	Average household occupancy	Numerical
Latitude	Geographic latitude of block group	Numerical
Longitude	Geographic longitude of block group	Numerical
MedHouseVal	Median house value in \$100k (TARGET)	Target

2. Exploratory Data Analysis (EDA)

2.1 Feature Distributions

The histogram grid below shows the distribution of all 9 features. Key observations: MedInc, AveRooms, AveBedrms, Population, and AveOccup are right-skewed, indicating a few high-value outliers. HouseAge is approximately uniform (houses of all ages are equally represented). The target variable MedHouseVal shows a spike at \$500k, which is a known data-capping artefact in the original 1990 census.

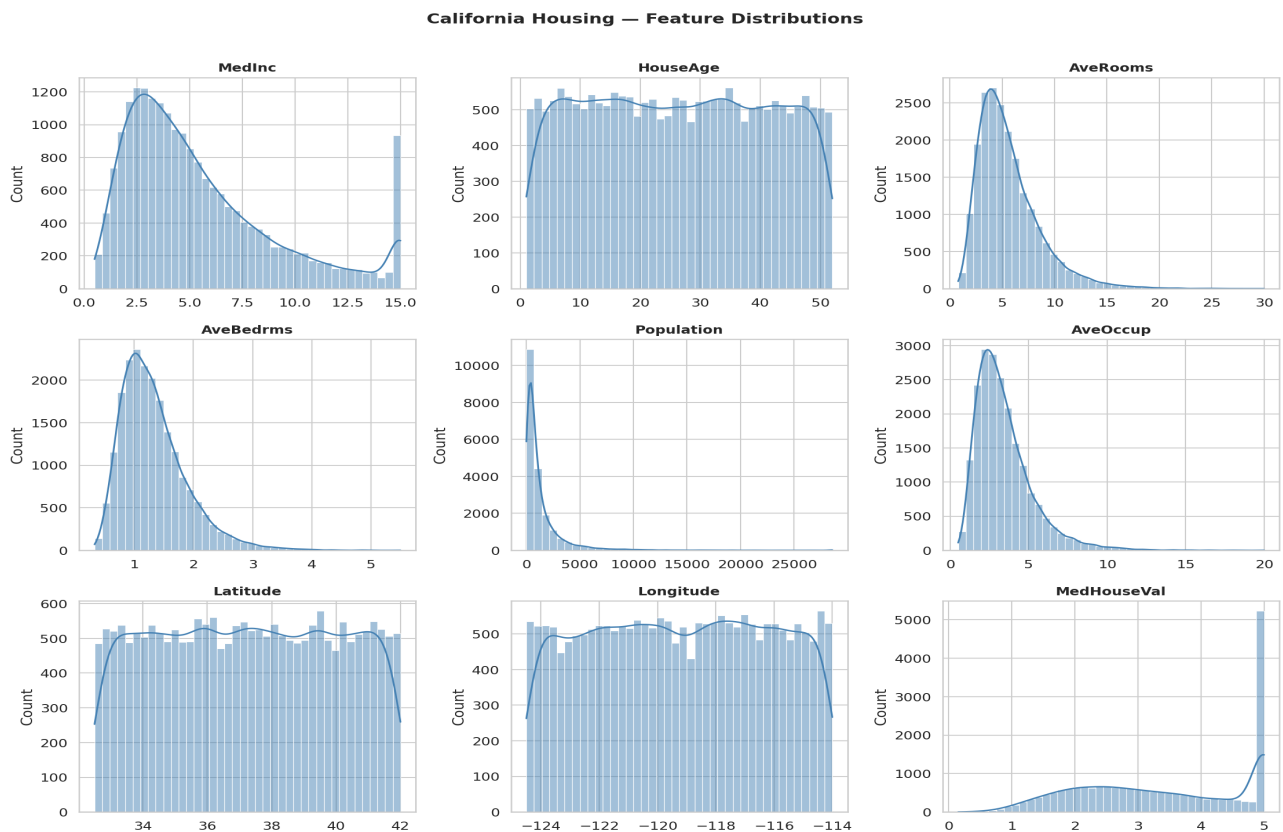


Figure 1: Distribution of all features in the California Housing dataset

2.2 Correlation Analysis

The heatmap below shows pairwise correlations between all features. The most important finding is that MedInc (median income) has a very strong positive correlation of 0.87 with MedHouseVal (house price). This makes intuitive sense — wealthier neighbourhoods have more expensive houses. All other features show very weak correlations with the target, suggesting that income is the dominant predictor in this dataset.

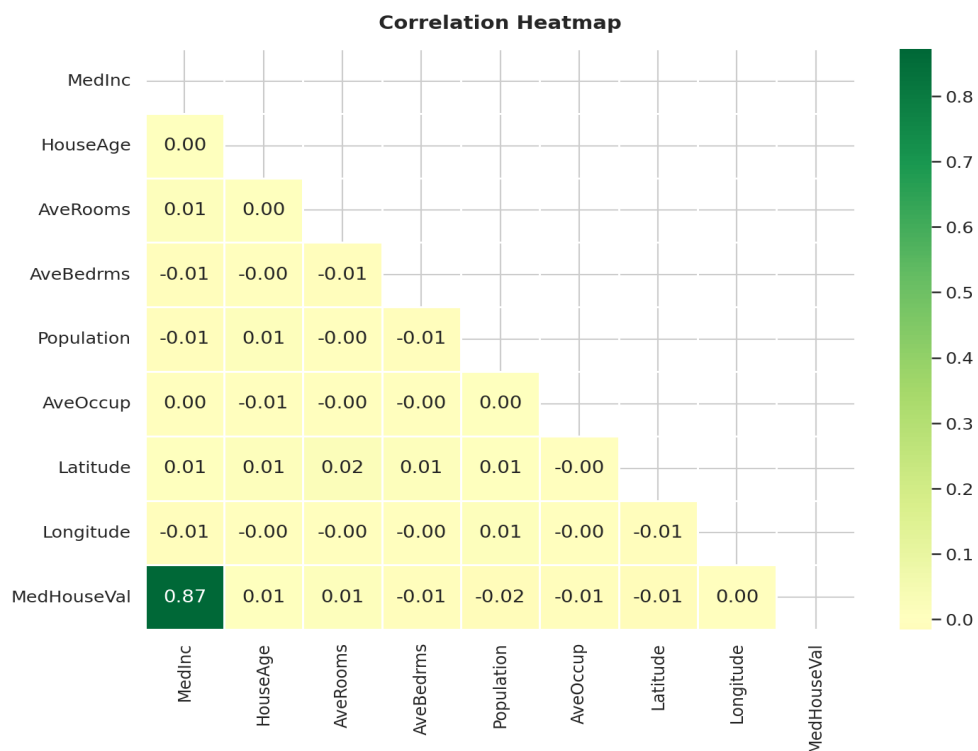


Figure 2: Correlation heatmap — MedInc shows 0.87 correlation with MedHouseVal

3. Data Preprocessing

Before training, the following preprocessing steps were applied:

- **Train/Test Split:** The dataset was divided into 80% training (16,512 samples) and 20% testing (4,128 samples) using `random_state=42` for reproducibility.
- **Feature Scaling:** `StandardScaler` was applied to normalize all features to mean=0 and standard deviation=1. This is essential for Linear Regression — without scaling, large-scale features like Population (range 0–28,000) would dominate over small-scale ones like AveBedrms (range 0.3–5).

4. Model Training

A Linear Regression model from `scikit-learn` was trained on the scaled training data. Linear Regression finds the best-fit linear relationship between the 8 input features and the target house price. The model learned that `MedInc` (income) is by far the most influential feature, with a standardised coefficient of +1.13.

5. Model Evaluation

The trained model was evaluated on the unseen test set of 4,128 samples using three standard regression metrics:

Metric	Value	Interpretation
MAE (Mean Absolute Error)	0.5248	On average, predictions are off by ~\$52,480
RMSE (Root Mean Squared Error)	0.6357	Penalises large errors more heavily than MAE
R2 Score	0.7598	The model explains 76% of variance in house prices

5.1 Actual vs Predicted Plot

The scatter plot below compares the model's predicted house values against the actual values from the test set. Points lying close to the red dashed diagonal indicate accurate predictions. The model performs well in the mid-range (\$150k–\$400k) and shows a known clustering effect at \$500k due to the data cap in the original dataset.

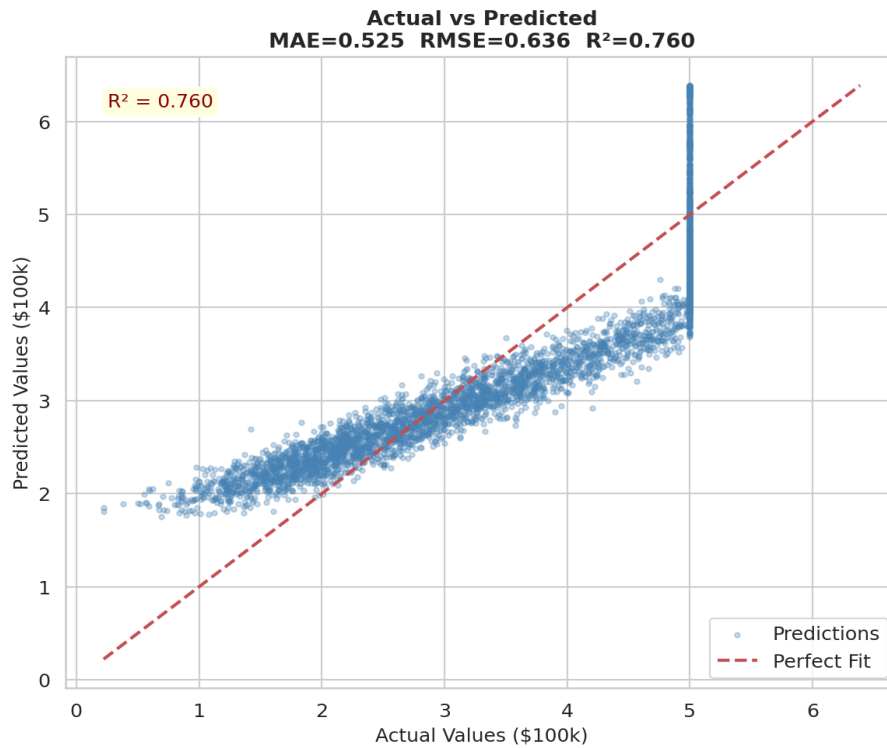


Figure 3: Actual vs Predicted scatter plot ($R^2 = 0.76$)

5.2 Residual Analysis

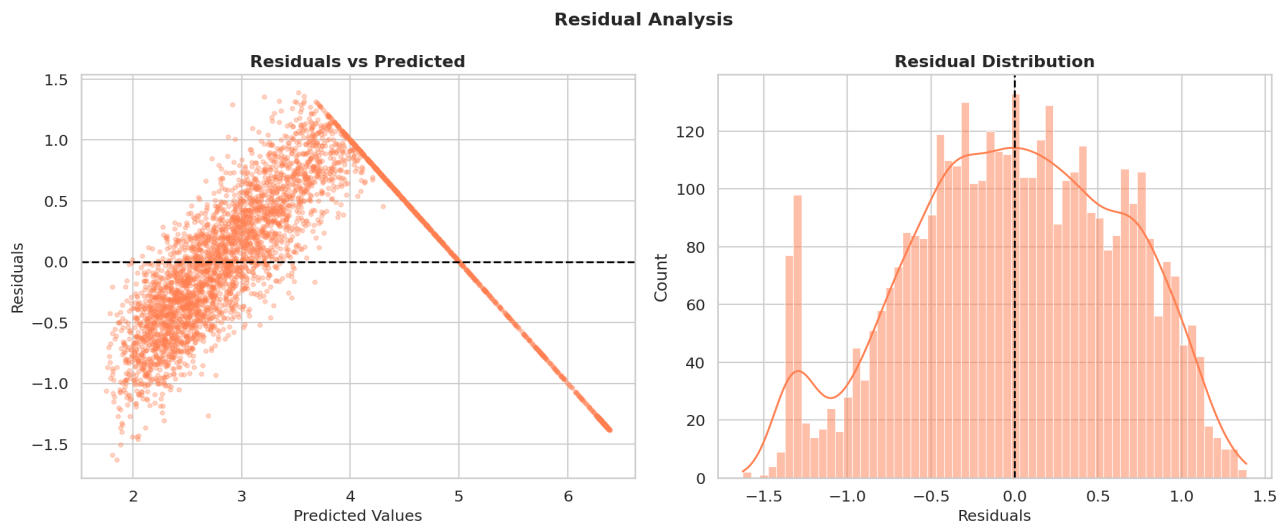


Figure 4: Residuals vs Predicted (left) and Residual distribution (right)

The residual plots show that errors are roughly centred around zero with an approximately normal distribution, which is a good sign. However, a slight fan-shaped pattern is visible in the left plot, suggesting mild heteroscedasticity — the model's errors grow larger for higher-priced houses. This is expected behaviour for a linear model on real estate data.

6. Feature Importance

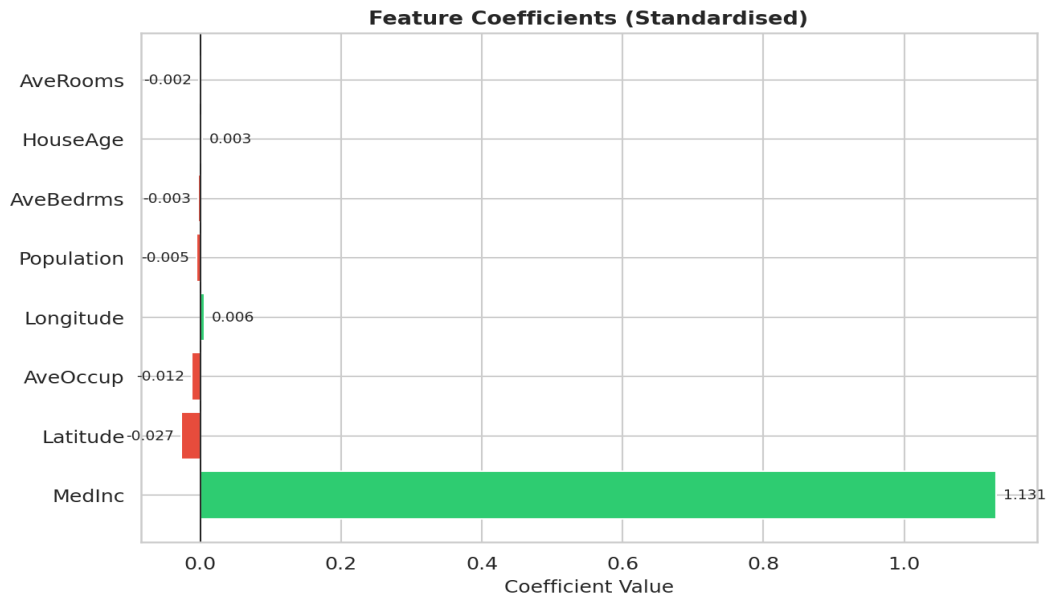


Figure 5: Standardised feature coefficients — green = positive effect, red = negative

MedInc dominates all other features with a coefficient of +1.13, confirming that median income is the strongest predictor. Latitude has a small negative effect (northern California tends to have lower prices). All other features have near-zero coefficients, indicating low individual predictive power in a linear model.

7. Conclusion

The Linear Regression model successfully learned to predict California house prices with an R2 score of 0.76, meaning it explains 76% of the variation in house prices. With an average prediction error of ~\$52,480, the model provides a solid baseline. The most critical insight is that median income is by far the strongest predictor of house price (correlation = 0.87), which aligns with real-world expectations.

8. Improvement Ideas

- **Ridge / Lasso Regression:** Add L1 or L2 regularization to reduce overfitting and handle multicollinearity between features.
- **Feature Engineering:** Create interaction terms like MedInc x Latitude, or log-transform right-skewed features to improve linearity.
- **Outlier Removal:** Remove the \$500k capped values and extreme Population outliers to improve model fit.
- **Non-linear Models:** Random Forest or Gradient Boosting (XGBoost) would capture non-linear patterns missed by linear regression and likely push R2 above 0.85.
- **Cross-validation:** Use k-fold cross-validation across the full dataset rather than a single train/test split for more robust evaluation.